LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of shaft axis, axis rotation, plane of rotation tip path plane, blade pitch, rotor thrust, relative air flow, coning angle, limit
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of shaft axis, axis rotation, plane of rotation tip path plane, blade pitch, rotor thrust, relative air flow, coning angle, limit of RPM.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of rotor diagram models; Safety: Handle models with care; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the main rotor?
(b) Define the term 'shaft axis' in the context of a helicopter.
(c) How does the rotor generate lift?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of shaft axis • Axis rotation mechanics • Plane of rotation and tip path plane • Blade pitch definition	L & Ds	PPT, IB	15 min
(b)	• Rotor thrust generation • Relative air flow dynamics • Coning angle explanation • RPM limit considerations	L & Ds	PPT, PDF	20 min
(c)	• Integration of rotor movements • Vibration checks basics • Adjustment procedures overview • Connection to flight controls	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how blade pitch affects rotor thrust.
(b) Describe the relationship between relative air flow and coning angle.
(c) Why is the limit of RPM critical for rotor safety?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of shaft axis and tip path plane relate to the components covered in the next lesson plan (LP 03).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Components of rotor mast, control system, swash plate, torque link, drag damper, droop restrainer, clutch mechanism, rotor brake, feathering sleeve etc.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min